

JAY SHAH

☎ +1 (817) 449-3968 · ✉ shahjay15102002@gmail.com · in linkedin/jay-shah · 🌐 github/jayshah13 · 🎓 M.S. CS · GPA 3.90 · May 2026

SUMMARY

Entry-level Data Scientist with M.S. Computer Science and hands-on experience building **predictive models**, conducting **statistical analysis**, running **A/B experiments**, and delivering **data-driven insights** to business stakeholders. Proficient in **Python**, **SQL**, **machine learning**, and **data visualization**. Experienced working end-to-end from raw data through model deployment, with strong written and verbal communication skills for translating complex findings into clear business recommendations.

WORK EXPERIENCE

University of Texas, Arlington

Aug 2024 – May 2026

Graduate Assistant — Data Science and Machine Learning

Arlington, TX

- 60+ research projects were producing model outputs with no evaluation standard – there was no way to measure whether models were actually working, catch when performance degraded, or present findings with statistical confidence to faculty and stakeholders.
- Built model evaluation frameworks in **Python (scikit-learn, Pandas, NumPy)**; applied **hypothesis testing**, **confidence intervals**, and **cross-validation**; created **Matplotlib + Plotly** visualizations; wrote clear written reports translating statistical findings into plain-language recommendations for both technical and non-technical audiences.
- 30% fewer errors**, **25% faster** evaluation cycles – every project had statistically defensible, clearly communicated findings backed by documented methodology.

Tata Consultancy Services

Jan 2024 – May 2024

Data Science Intern

Gandhinagar, India

- 500K+ operational records** had no ML-driven decision support – manual processes were slow, inconsistent, and a hidden data quality issue had been silently skewing results for weeks without anyone noticing.
- Framed business problems as ML tasks; built and tuned **classification and regression models** using **XGBoost**, **LightGBM**, and **scikit-learn** with **Python + Pandas**; ran **A/B tests** to validate model impact; deployed models into live workflows; communicated findings and performance updates regularly to technical and business stakeholders.
- 2 models in production**, system **30% more reliable**, delivery **15% faster** – data-driven decisions replaced manual guesswork with statistically validated, clearly explained recommendations.

Celebal Technologies

May 2023 – Sept 2023

Data Analyst and BI Intern

Jaipur, India

- Three business teams measured the same KPIs differently – **1M+ records** had inconsistencies nobody had caught, and every leadership meeting started with debates about whose numbers were right instead of decisions.
- Cleaned and analyzed data using **SQL + Python (Pandas)**; applied **statistical methods** including outlier detection, distribution analysis, and **hypothesis testing**; built **Power BI** and **Tableau** dashboards; presented findings and business recommendations to leadership across teams.
- Data accuracy **18% higher**, reporting workload down **25%** – unified, statistically sound data gave every team a single trusted foundation for decision-making.

PROJECTS

Customer Risk and Behavior Prediction Model · Python · XGBoost · LightGBM · scikit-learn · Pandas · SQL · Plotly Dec 2025 – Mar 2026

- A business had no way to predict which customers were at risk of churning, defaulting, or disengaging – every intervention came too late because there was no model to surface warning signals early.
- Collected and prepared **300K+ behavioral records** using **SQL + Pandas**; performed **exploratory data analysis** to identify patterns; engineered features; trained and compared **XGBoost** and **LightGBM** classifiers with **cross-validation**; validated business impact with **A/B testing**; built **Plotly** dashboards presenting risk scores and recommendations to non-technical stakeholders.
- Model achieved **85% ROC-AUC**, manual analysis reduced **40%** – business teams could identify and act on at-risk customers weeks before problems occurred, with confidence in the statistical evidence behind every recommendation.

Automated Early Warning and Anomaly Detection System · Python · scikit-learn · PyTorch · SQL · statistical modeling · Tableau Apr – Jul 2025

- Operations and product teams had no systematic way to detect when something was going wrong – problems were only discovered after they had already caused damage, and findings were never documented in a way others could act on.
- Performed **EDA** to understand normal vs. anomalous patterns across **12K+ records**; built **statistical and deep learning anomaly detection models** using **scikit-learn** and **PyTorch**; implemented **hypothesis testing** to confirm statistical significance; designed **Tableau** dashboards with written narrative summaries explaining findings and recommended actions.
- 90% of issues** caught automatically before causing problems – teams received clear, plain-language alerts with statistical confidence levels they could act on immediately.

AI-Powered Insight and Forecasting Engine · Python · PyTorch · LangChain · OpenAI APIs · SQL · Apache Airflow · AWS Mar – May 2024

- 1M+ records** sat unused because neither data scientists nor business users had a fast, reliable way to extract insights – ad hoc queries took days and forecasting models were built once and never updated.
- Built a **forecasting pipeline** using **time-series models** and **PyTorch** for trend prediction; integrated **LangChain + OpenAI APIs** to enable natural language querying of data; orchestrated with **Apache Airflow** on **AWS**; produced automated insight reports and **Power BI** dashboards updated on a regular cadence.
- Forecasting accuracy improved **30%**, manual analysis time reduced **35%** – business teams could ask questions in plain English and receive data-backed answers automatically, without waiting for analyst support.

EDUCATION

University of Texas, Arlington

Aug 2024 – May 2026

M.S. Computer Science · GPA: 3.90/4.00 · Machine Learning · Statistics · Data Mining · Deep Learning · Algorithms

Arlington, TX

Charusat University

Oct 2020 – May 2024

B.S. Computer Engineering · GPA: 3.78/4.00 · Probability & Statistics · Database Systems · Business Analytics

India

SKILLS

Languages: Python (advanced) · SQL (advanced) · R (familiar) · Scala (familiar) · Bash

Machine Learning: scikit-learn · XGBoost · LightGBM · Random Forest · SVM · regression · classification · clustering · anomaly detection

Deep Learning: PyTorch · TensorFlow (familiar) · CNNs · RNNs · Transformers · NLP · computer vision · model fine-tuning

Generative AI: LangChain · OpenAI APIs · Anthropic APIs · RAG · prompt engineering · LLM integration · embeddings

Statistics: Hypothesis testing · A/B testing · confidence intervals · p-values · regression analysis · causal inference · experimental design · Bayesian methods

Data Wrangling: Pandas · NumPy · PySpark · SQL joins/aggregations · feature engineering · data cleaning · EDA · time-series analysis

Visualization: Tableau · Power BI · Plotly · Matplotlib · Seaborn · Looker (familiar) · executive dashboards · data storytelling

ML Lifecycle: Model training · cross-validation · hyperparameter tuning · model monitoring · out-of-sample testing · deployment · MLflow (familiar)

Cloud & Infrastructure: AWS (S3, EC2, SageMaker) · GCP (BigQuery, Vertex AI – familiar) · Azure ML (familiar) · Docker · Git · Apache Airflow

Databases: PostgreSQL · MySQL · MongoDB · Snowflake · Google BigQuery · DynamoDB · data warehouse design

Communication: Written reports · executive presentations · data storytelling · translating findings for non-technical audiences · cross-functional collaboration